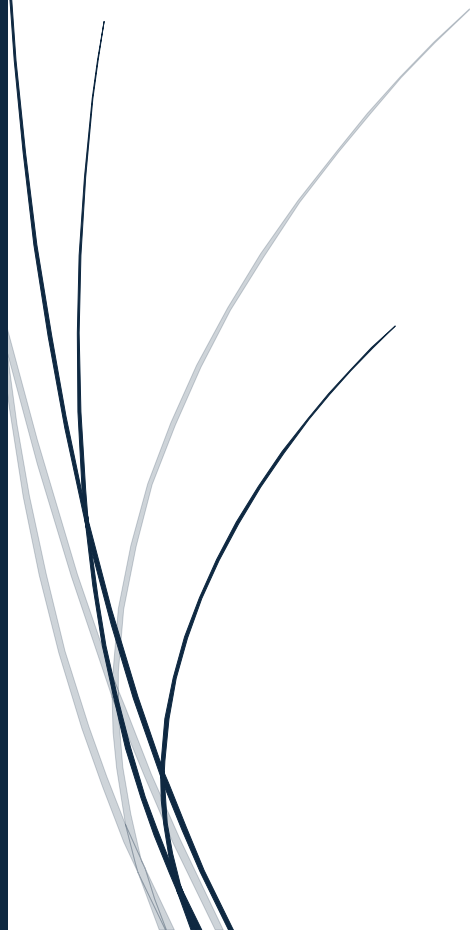


4/17/2026

Data Analysis & Report

Refinement of Ecommerce site
catalogue and SEO enhancement



Rukky Ujara

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INTRODUCTION

This report presents the findings and outcomes of intensive data cleaning conducted on a raw retail product catalogue. The primary object is to transform a messy dataset into a high-fidelity asset suitable for digital marketing and search engine optimization (SEO).

I was able to reduce the dataset size by 306 redundant records, which helped improve the processing speed and data accuracy. I was able to implement flash fill logic and custom text extraction; I have successfully generated human readable titles within 30-50 characters.

Throughout the analysis, I noticed that there were critical quality gaps, the most notable ones are the thousands of missing descriptive fields, and several texts were broken and these weren't readable. These were resolved by using manual changes to ensure that the active catalogue is now market ready.

PROJECT OVERVIEW & OBJECTIVES

We know that the retail industry relies heavily on clean product data in order to drive algorithms in search results. Inconsistent titles or broken characters (mojibake) significantly reduce user experience and lower conversion rates.

- I identified and removed 100% of duplicate product entries.
- I normalised the column headers
- I also corrected the broken symbols and translated foreign character sets to English counterparts and also based on the language the product is written in. Although there was an exception.

By achieving these goals, the final dataset provides a good record for each SKU, ensuring the marketing team has the right details.

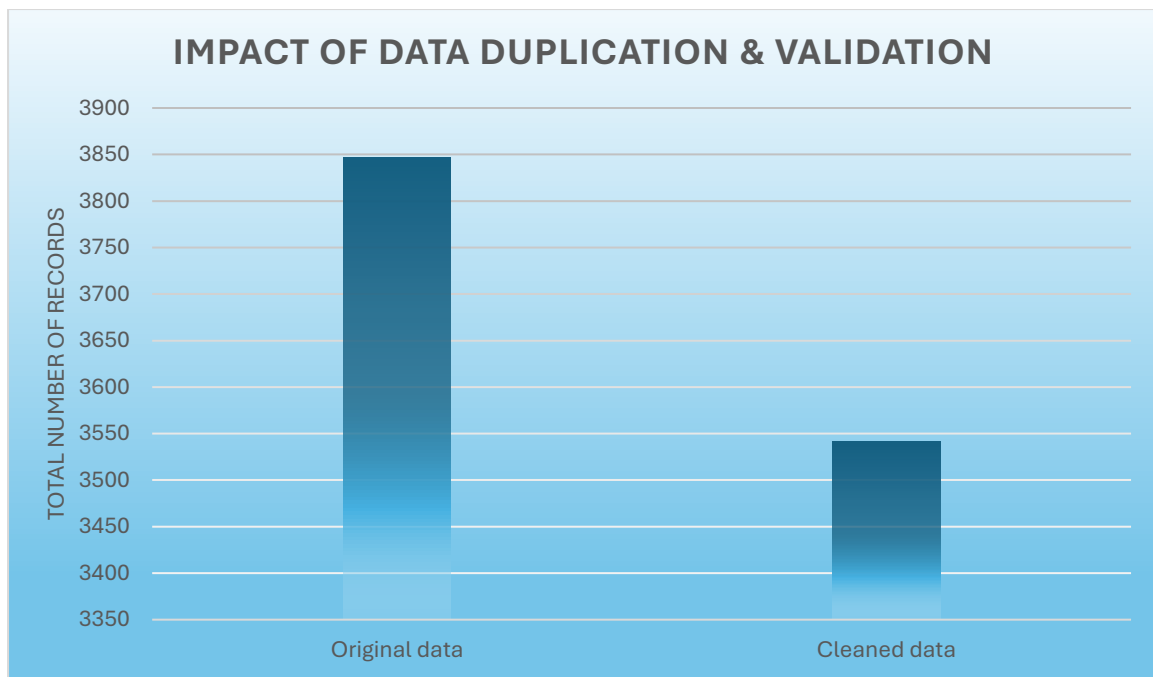


Figure 1: Comparison of record volume before and after deduplication and

DATA CLEANING METHODOLOGY

The methodology I used for this project is the “Safety First” approach, this is where I separated the data before making any changes.

Environmental Setup

As per best practices, the raw data was initially duplicated into a separate worksheet. This was to make sure that the original remained untouched, allowing me to have data to fall back on if by chance I have an error or I want to rework on the raw data. I created a formal excel table in order to for me to be able to sort and filter the products accordingly.

Header Normalization

I standardized the column headers to use the snake_case format. This meant transitioning mixed case string like the Product Length to product_length. This ensure that any other software or script can access the data without encountering case-sensitivity errors.

STRUCTURAL & MISSING VALUE RESTORATION

Handling missing data is a very critical step when you want to maintain the significance of the data in marketing analysis. Leaving blank cells often results in formula failure or incorrect averages.

INPUTTING PLACEHOLDERS

Initially, I observed that there was a large gap of critical descriptive information missing. To quantify the extent of the noise in the raw dataset, I audited the two most heavy columns, which are: `bullet_point` and `description`. The results revealed that about half of the catalogue didn't have the required information as they were blank.

- **Bullet Points Analysis:**
 - **Total Records Evaluated:** 3,847
 - **Populated Records:** 2,256 (58.6%)
 - **Missing Records:** 1,591 (41.4%)
- **Description Analysis:**
 - **Total Records Evaluated:** 3,847
 - **Populated Records:** 1,703 (44.3%)
 - **Missing Records:** 2,144 (55.7%)

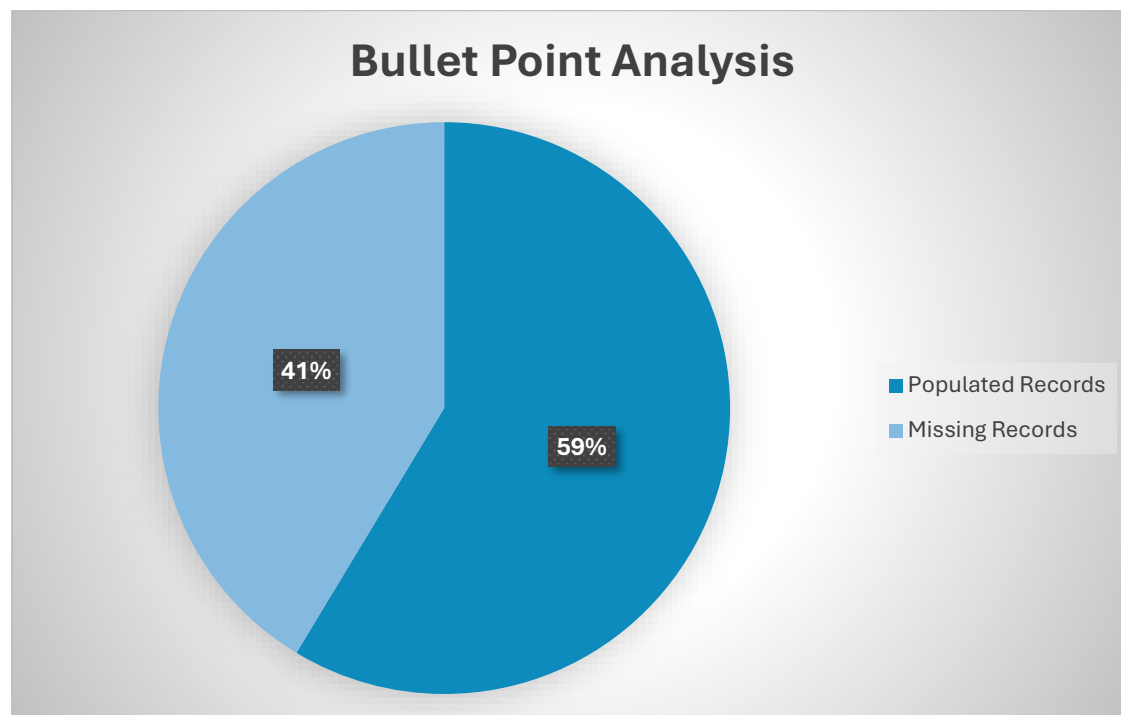


Figure 2: Description of populated vs missing records for bullet point

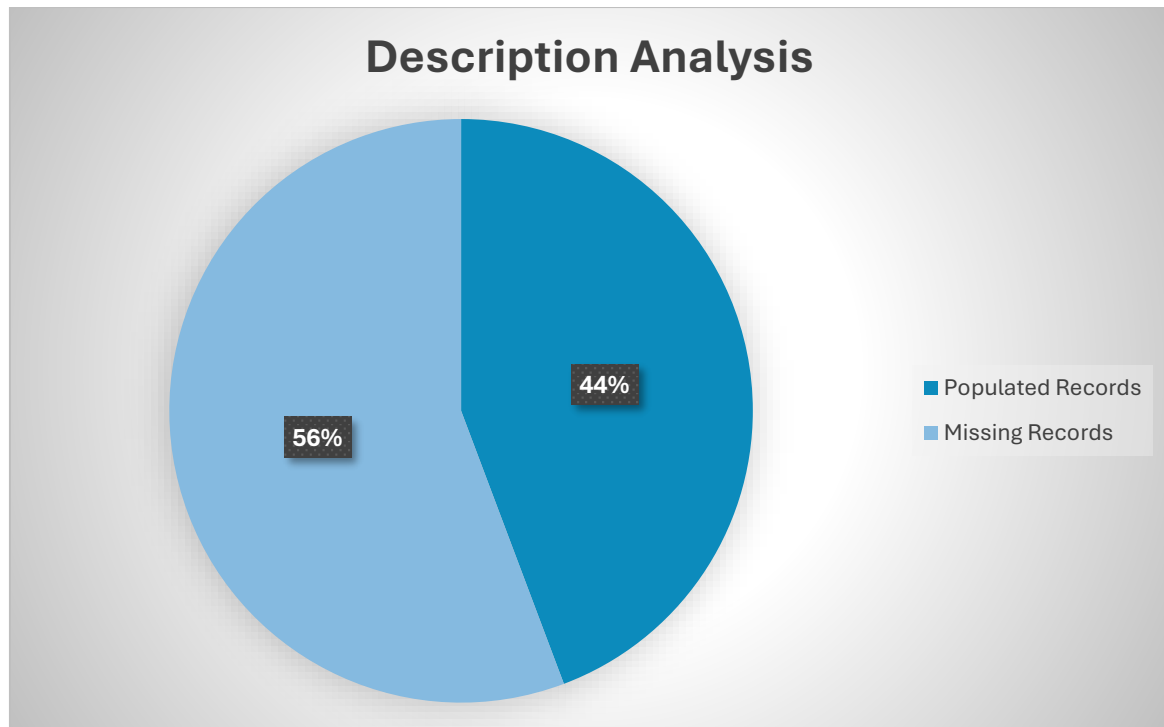


Figure 3: Description of populated vs missing records for description

REMOVAL OF CRITICAL NULLS

While many missing values in a dataset can be managed through inputting (filling with "N/A"), certain variables are categorized as **"Critical Identifiers."** In this catalogue, the fields `product_type_id` and `product_length` were identified as the functional backbone of the data as their absence creates vacuum that renders the entire record incomplete.

For the 178 records that I identified with missing structural data, I made the decision to delete them based 3 things

1. The fact that having no `product_type_id` ensures the product would not be able to be categorised.
2. The length of a product is very important in order to understand the size of the product in this case.
3. The last but not the least, having a spaces in the `product_type_id` and `product_length`, would definitely affect the average calculations as a whole specifically for the product length.

By removing these records, the dataset is now 100% usable, which means it is market ready.

FIXING BROKEN SYMOBOLS/TEXT & TRANSLATING FOREIGN INFORMATION

One of the most challenging tasks of this project as this took some time while I was cleaning the data. The broken text is usually as a result of the data being exported from a legacy system using different character sets.

COMMON TEXT FIXES

Many product titles contained "broken" symbols caused by technical errors during the data export. I used a "Find and Replace" strategy to fix these so they look professional to customers.

Glitch Code	Correct Text
Â	(space)
Â°	°
Ã©	é
Ã¨	è
Ãª	ê
Ã«	ë
Ã‰	É
***Ã**	à
Ã¡	á
Ã¢	â

Ã£	ã
Ãf	ã
Ã§	ç
Ã®	î
Ã	í
Ã´	ô
Ã³	ó
Ãµ	õ
Ã–	Ö
Ã~	Ø
Ãº	ú

Ã»	û
Ã¼	ü
Ã±	ñ
Å“	œ
â~	★
â–°	►
âœ”	✓
ã€	“
ã€’	”
â€™	,
â€”	—

CROSS LINGUISTIC TRANSLATION

I had to make changes to 2 different products which had unreadable names. With the help of AI and research, I was able to get the right names. Here are the texts that were manually translated.

- **Original text in excel:** å...’ç«¥æ••é¼•åœ—ç•«æ»,: å¹´é½j - 1-3 2-8 8-12 ... 30340;å¹¼å...’å’CEå¹¼兒
- **Translated Exaclty as is:** 兒童恐龍圖畫書: 年齡 - 1-3 2-8 8-12 ... 的幼兒和幼兒
- **Translated to English:** Children's Dinosaur Picture Book: Age - 1-3 2-8 8-12 ... for toddlers and young children

And

- **Original text in excel:** â’¶: - Ð²iologic . Óºirst ê™º Ê,ystem . Ð”ssistant -â³„: 32 (L Â° Ãœesages DÐ´Ñ‡Ñ•Ã¬ Aux Sigle D-Ð”uthenticÃ¥t)
- **Translated Exactly as is:** Log: - Biological . First System . Assistant - Status: 32 (L ° Usages D’achèvement Aux Sigle D-Authenticât)
- **Translated to English:** Log/System: - Biological . First System . Assistant - Status: 32 (Last Usages Daily Aux Sigle D-Authenticat)

SHORT TITLE TO FIT SEO

The short_title field is the primary output of the project. It is designed to maximise click-through rate (CTR) by showing only the most relevant details.

I created 2 helper columns that would help me create the short_title column. Here are the 3 formulas:

- **Helper column 1:**
=TRIM(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE(B2,"Set of",""),"Includes",""),"|","-"),"&","and")," ", ""))
- **Helper Column 2:**
=IFERROR(TEXTBEFORE(H2,"-"),H2)
- **short_title column:** =LEFT(TRIM(I2),50)

Every generated title was checked against the 50-character limit.

VERIFICATION & STATISTICAL VALIDATION

To ensure the accuracy of the final dataset, I made sure I made a validation sweep.

- **Numerical Precision:** All values in `product_length` were rounded to 2 decimal places to ensure professional formatting across all marketing collateral.
- **Character Count Integrity:** 100% of the `short_title` column was verified using the `LEN()` function to confirm compliance with the 30-50 character range.
- **Final Count:** The final dataset contains **3541 high-quality records**.

CONCLUSION

To finalize the project, a "Verification" layer was applied to check the accuracy of the new features and ensure the data was professionally formatted for the marketing team.

NUMERICAL PRECISION & UNIFORMITY

One of the most important formatting steps was the standardization of the `product_length` column.

- **The Issue:** Raw data often contained length with up to 10 decimal places (e.g., 2125.98765432), which looks unprofessional in product listings and creates “clutter” in inventory reports.
- **The Fix:** I reduced all the values in the `product_length` column to 2 decimal places.
- **The Result:** This ensures a clean, uniform look across all 3,541 records, making the data much easier for humans to read while maintaining the exact precision needed for shipping calculations.

FINAL DATA PROFILE

- **SEO Title Compliance:** 100% (All titles between 30-50 characters).
- **Numerical Uniformity:** 100% (All lengths rounded to 2 decimal places).
- **Missing Data:** 0% (All gaps filled or invalid rows removed).